Important Announcement About Laboratory

- A Google Classroom is opened for our laboratory experiments. Please join the classroom using the link below (you need to have a Google Account):

  https://classroom.google.com/c/NjI5MDEwNzEwOTcw?cjc=g6umm7z

- Based on the day of the experiment you entered the following Experiment 1 report submission links with their related deadlines are available in the Google Classroom:
  - EXPERIMENT 1 - **27 OCTOBER Thursday Groups** Report Submission
  - EXPERIMENT 1 - **28 OCTOBER Friday Groups** Report Submission
  - EXPERIMENT 1 - **02 NOVEMBER Thursday Groups** Report Submission
  - EXPERIMENT 1 - **03 NOVEMBER Friday Groups** Report Submission
- Please add the additional template (given in the link below) to your report, listing group member names, surnames, student numbers, signatures and each member's specific individual contributions to the report preparation:

  https://docs.google.com/document/d/1HR0a3Zmv3dL5xoBZpf4v5PcKgvwXsxf8/edit?usp=drive_link&ouid=113193094559479887472&rtpof=true&sd=true

Please go to the next page

Important Announcement About Laboratory

- The reports **have to be prepared by handwriting**. If specific figures, graphs and screenshots are particularly needed in some experiments, these **can be added as extra pages to your handwritten report.**
- Every group should **submit one group report per experiment**. Please **do not submit several copies of the same report** by different group members. Maximum two people should submit from one group.

EE2001 Electrical Circuits- I:

Continuation to:

Techniques of Circuit Analysis

31 October 2023

Outline



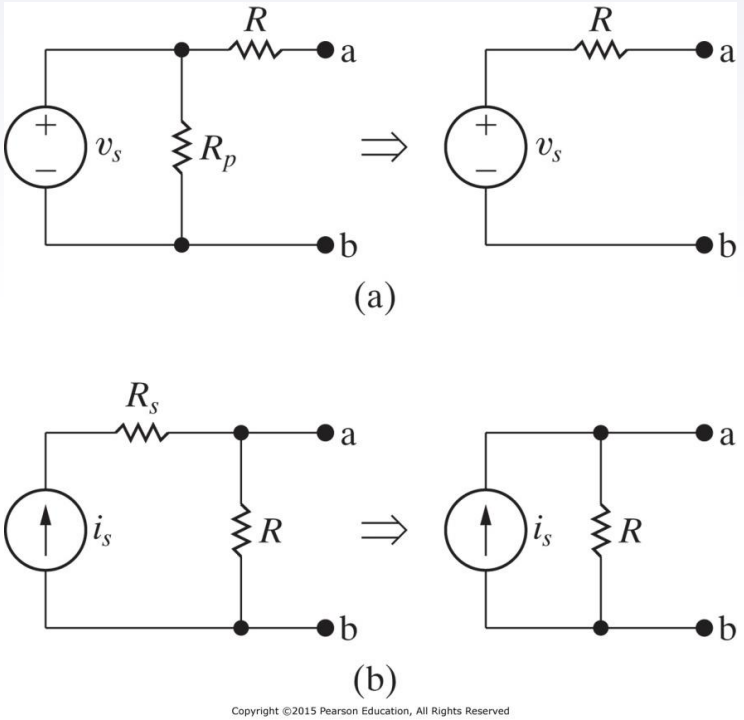
Continuation to: 10. Source Transformations

11. Thevenin and Norton Equivalents

12. Maximum Power Transfer

9 Source Transformations

- Resistances in **parallel to a voltage source** or in **series to a current source do not affect** the source transformation.
- Such resistors **do not have effect on** the equivalent circuit. Therefore, they **<u>can be removed</u> during** source transformation.

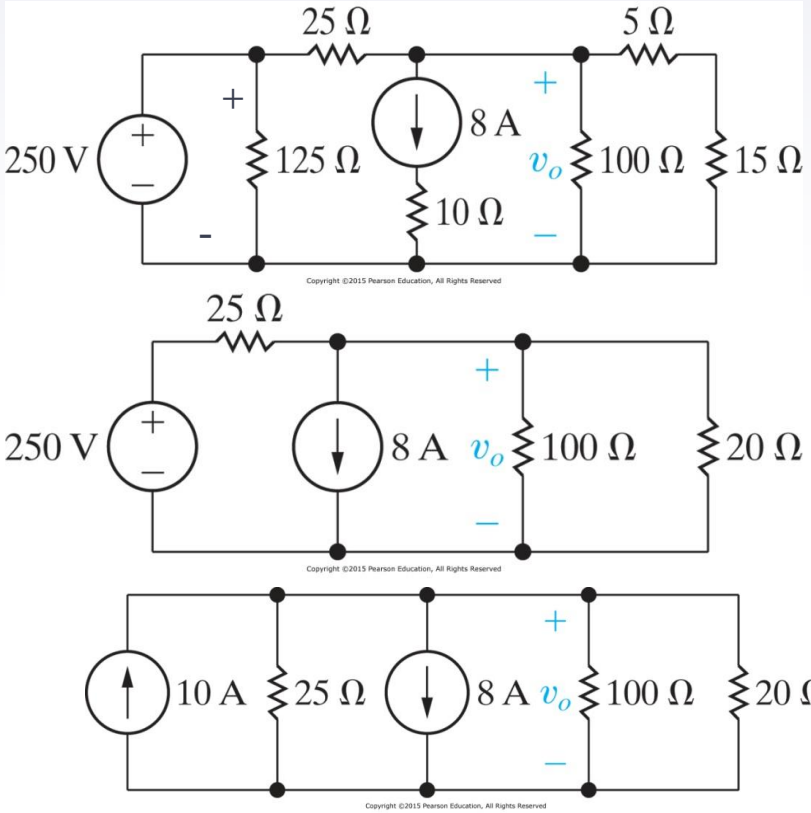


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Source Transformations

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- Example-2: a) Use source transformations to **find the voltage v_o** in the circuit.



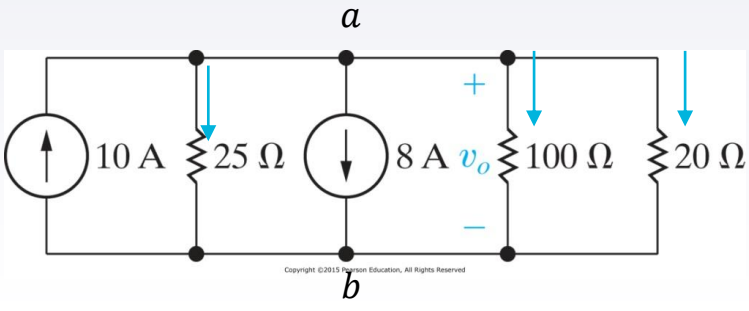
a) We begin by **removing** the 125 Ω resistor connected across the 250 V voltage source and the 10 Ω resistor connected in series with the 8 A current source.

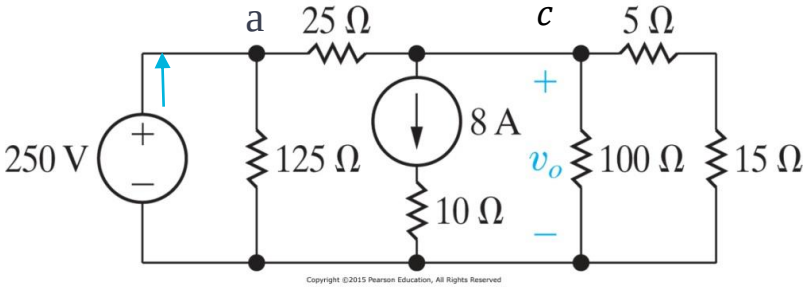
- We also **combine** the series-connected resistors into a single resistance of 20 Ω.
- We now use a source transformation to **replace** the 250 V source and 25 Ω resistor with a 10 A current source in parallel with the 25 Ω resistor.

Source Transformations

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- Example-2:





- The circuit now can be **simplified** by using **Kirchoff's Current Law** to **combine** the parallel current sources into a single source. Also, the parallel resistors **combine** into a single resistor. These can be done since all the components are connected **to the the same** upper node and the lower node.

$$\frac{1}{R_{eq}} = \frac{1}{25} + \frac{1}{100} + \frac{1}{20} = \frac{4+1+5}{100} = \frac{10}{100} \qquad R_{eq} = \frac{100}{10} = 10\ \Omega$$

- Using the simplified circuit,

$$v_o = (10)(2) = 20\ V$$

b) Find the **power developed by the 250 V source**.

- Considering the current of the 250 V source as i_s and writing KCL equation for node a,

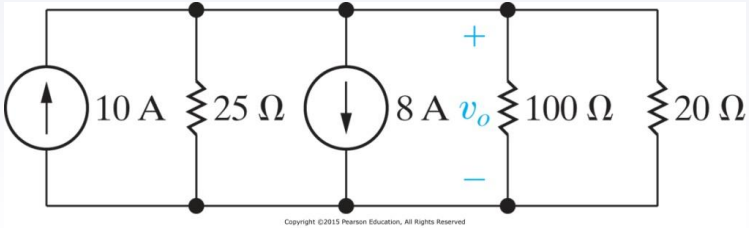
$$-i_s + \frac{250}{125} + \frac{250-20}{25} = 0 \qquad i_s = 11.2\ A$$

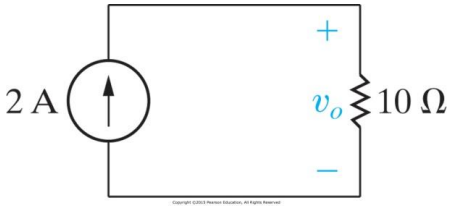
$$p_{250V} = -(250)(11.2) = -2800\ W$$

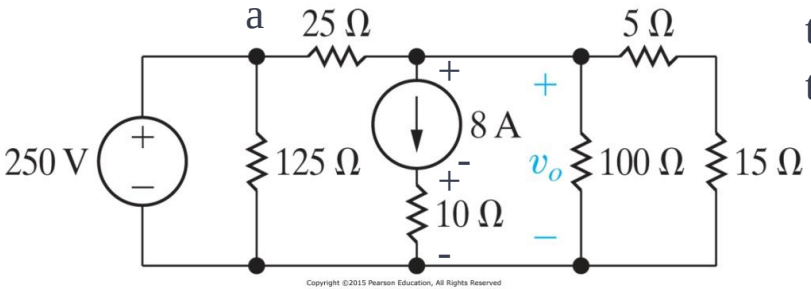
Source Transformations

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- Example-2:







c) Find the **power developed by the 8 A current source**.

- We first need the voltage across the 8 A current source. Writing **KVL equation** for the **mesh** with 8 A current source, 10 Ω resistor and 100 Ω resistor,

$$-v_s - 8(10) + v_o = 0 \qquad v_s = 20 - 80$$

$$v_s = -60\ V$$

$$p_{8\ A} = (-60)(8) = -480\ W$$

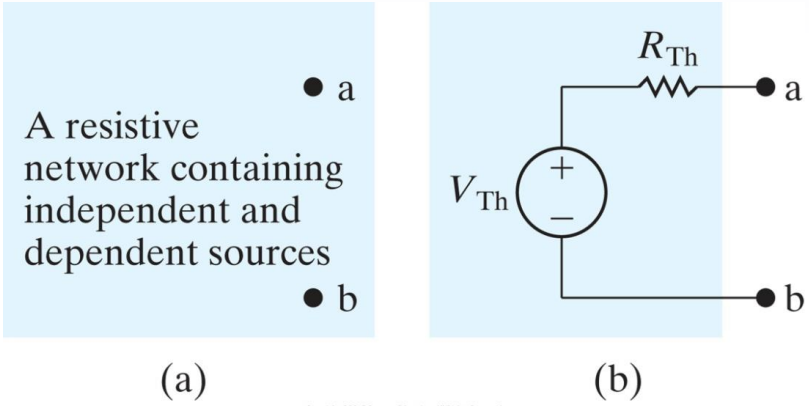
Note that the 125 Ω and 10 Ω resistors **do not affect the value of v_o, <u>but do affect the power calculations</u>**. We **neglected** them **during source transformations**; **but** we **considered** them again, **when making power calculations**.

10 Thevenin and Norton Equivalents

- The Thevenin and the Norton equivalents are **circuit simplification** techniques that focus on the **terminal behavior** of circuits.
- They can be used in circuits composed of **linear elements**.

The Thevenin Equivalent

- Consider any circuit composed of sources (both dependent and independent) and resistors.
- The **Thevenin equivalent** is equivalent to the original circuit in the sense that if the same load resistor is connected to both networks/circuits **the current and voltage in both circuits are the same**.

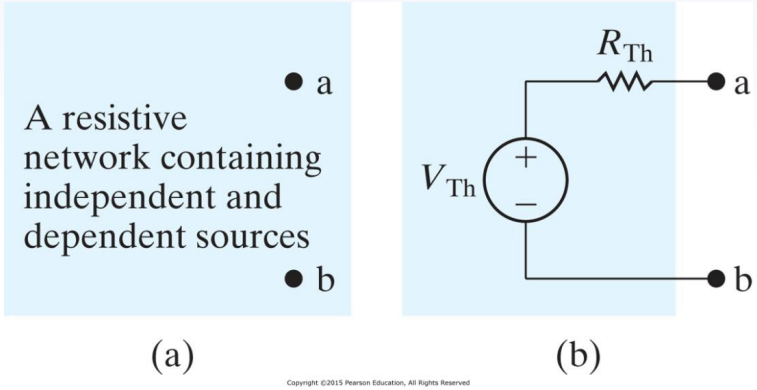


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- In other words, **their terminal current-voltage characteristics are the same**.
- The equivalence holds **for all possible values of the load resistance**.

The Thevenin Equivalent

- In order to determine the **Thevenin equivalent** of a given circuit, we need to determine $\boldsymbol{V_{Th}}$ and $\boldsymbol{R_{Th}}$



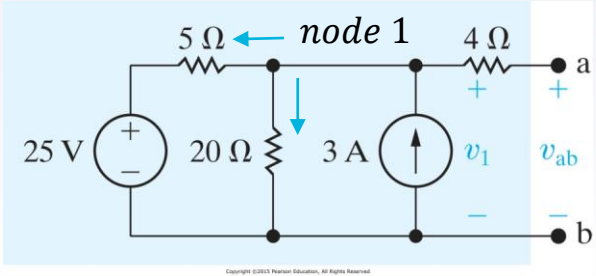
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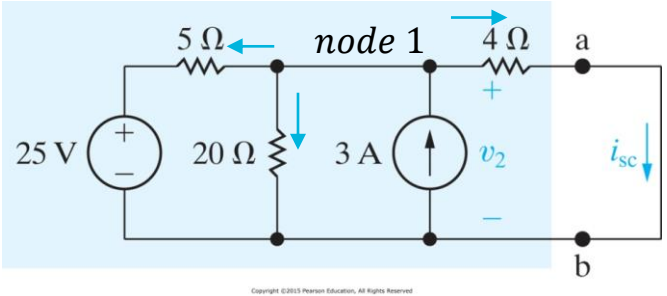
- Considering the **open circuit** and **short circuit conditions**, we can derive:
  - $V_{Th} = V_{Oc}$ =Open circuit voltage
  - $R_{Th} = \frac{V_{Oc}}{I_{Sc}}$ = Open circuit voltage/Short circuit current

The Thevenin Equivalent

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- Example:





- We first calculate the **open-circuit** voltage of v_{ab}
- When the terminals a nd b are **open**, there is **no current** in the 4 Ω resistor. Therefore the open circuit voltage v_{ab} is **identical** to the voltage across the 3 A current source.
- By solving **a node-voltage equation**, choosing the lower node as the reference node,

$$\frac{v_1-25}{5}+\frac{v_1}{20}-3=0 \qquad v_1=32\,\text{V}=V_{Th}$$

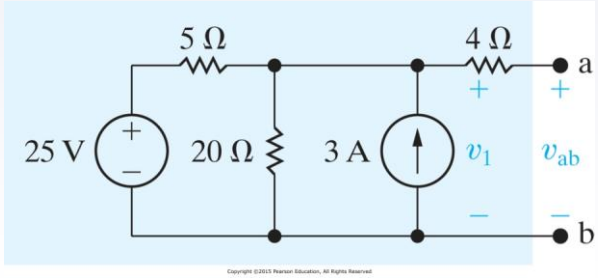
- The next step is to place **a short circuit** across the terminals and calculate the resulting short-circuit current. Short-circuit current **is in the direction** of the open-circuit voltage drop across the terminals a, b.
- Once v_2 is **known**, i_{sc} **can be found** easily. Again using the lower node as the reference node,

$$\frac{v_2-25}{5}+\frac{v_2}{20}-3+\frac{v_2}{4}=0 \qquad v_2=16\,\text{V} \qquad i_{sc}=\frac{16}{4}=4\,A$$

The Thevenin Equivalent

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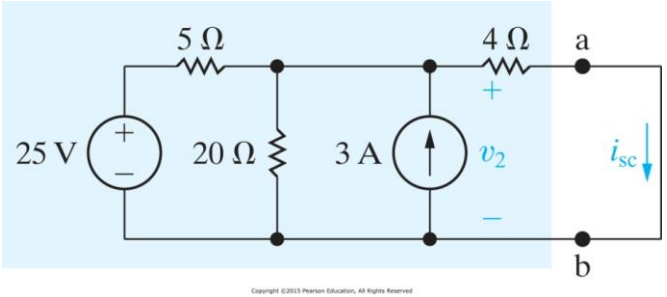
- Example:

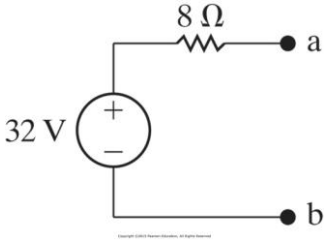


$$V_{Th} = 32\ V$$

$$i_{sc} = 4\ A$$

$$R_{Th} = \frac{V_{Th}}{i_{Sc}} = \frac{32}{4} = 8\ \Omega$$





End of the Lesson (31.10.23)

Any questions?

Supplementary videos: Enter Youtube-> Search for ‘Alparslan Zehir’ -> Go to Playlists -> Watch ‘Electrical Circuits’ playlist